

Mid Semester - Winter Examination (2025-2026)

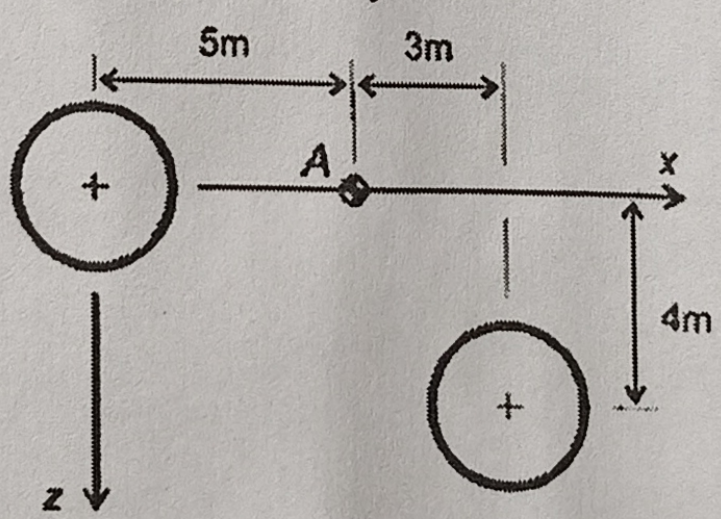
VIII Semester B.Tech. (Mining Engineering)

Underground Space Technology (MND407)

Time: 1 Hour

Maximum Marks: 30

Instructions: 1) Answer ALL questions 2) Assume suitable data wherever required

Q.No	Question	Marks														
1	During an underground site investigation, core samples were obtained from exploratory drilling. The borehole depth intervals and corresponding lengths of core pieces recovered in each run are listed below: <table border="1"> <thead> <tr> <th>Borehole run (m)</th> <th>Length of the core recovered (cm)</th> </tr> </thead> <tbody> <tr> <td>0.0-1.0</td> <td>12, 2, 13, 24, 5, 10, 9</td> </tr> <tr> <td>1.0-2.0</td> <td>24, 15, 6, 15, 9.75, 9.85, 10</td> </tr> <tr> <td>2.0-3.0</td> <td>20, 15, 25, 5, 15, 6, 9, 2</td> </tr> <tr> <td>3.0-5.0</td> <td>25, 24, 28, 12, 15, 18, 12, 15, 4, 5, 6, 7, 8, 9, 12</td> </tr> <tr> <td>5.0-7.0</td> <td>15, 14, 12, 13, 5, 8, 7, 2, 8, 9</td> </tr> <tr> <td>7.0-10.0</td> <td>15, 14, 16, 17, 2, 5, 8, 9, 12, 15, 14, 18</td> </tr> </tbody> </table> The rock mass parameters obtained from the site investigation are: $J_n = 8$, $J_r = 1.3$, $J_a = 2.0$, $J_w = 1.0$, $SRF = 2.3$. Assume that the average uniaxial compressive strength (UCS) of the intact rock is 100 MPa. a) Determine the Rock Mass Quality Index (Q-value) of the rock mass. b) Estimate the mean deformation modulus of the rock mass using the Q-system relationship.	Borehole run (m)	Length of the core recovered (cm)	0.0-1.0	12, 2, 13, 24, 5, 10, 9	1.0-2.0	24, 15, 6, 15, 9.75, 9.85, 10	2.0-3.0	20, 15, 25, 5, 15, 6, 9, 2	3.0-5.0	25, 24, 28, 12, 15, 18, 12, 15, 4, 5, 6, 7, 8, 9, 12	5.0-7.0	15, 14, 12, 13, 5, 8, 7, 2, 8, 9	7.0-10.0	15, 14, 16, 17, 2, 5, 8, 9, 12, 15, 14, 18	6+2=8
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2	The figure given below illustrates the relative positions of two parallel, horizontal circular tunnels, each having a diameter of 3 m. Before excavation, the rock mass is subjected to a uniform in-situ stress field such that: $P_{xx} = P_{yy} = P_{zz} = 11$ MPa. a) Estimate the approximate values of stress distribution along the boundaries of both tunnel excavations after excavation. b) Determine the principal stresses acting at point A, as indicated in the figure. Clearly state any assumptions made in your analysis. 	5+3=8														
3	Briefly explain the different geological and geotechnical investigations required for constructing underground rock caverns in a hydroelectric project, and prepare a structured data sheet summarizing the details of these investigations.	8														
4	Describe briefly the step-by-step procedure for solving rock engineering design problems using a flowchart, integrating the key design principles.	6														

Note: Use the following formulas wherever required

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$$\sigma_{rr} = \frac{p}{2} \left[(1+K) \left(1 - \frac{a^2}{r^2} \right) - (1-K) \left(1 - 4\frac{a^2}{r^2} + \frac{3a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{\theta\theta} = \frac{p}{2} \left[(1+K) \left(1 + \frac{a^2}{r^2} \right) + (1-K) \left(1 + \frac{3a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{r\theta} = \frac{p}{2} \left[(1-K) \left(1 + \frac{2a^2}{r^2} - \frac{3a^4}{r^4} \right) \sin 2\theta \right]$$